

# Arch-I-Scan: Archiving Our Biggest Digital Collection Yet

Scaling our workflow to deal with an unusually large deposit

Marco Brunello | marco.brunello@york.ac.uk | ORCID 0009-0003-5405-3861

## WHAT IS ARCH-I-SCAN?



A project directed by Prof. Penelope Allison and Prof. Ivan Tyukin at the University of Leicester including photographs and 3D models of ceramic remains from Roman Britain - sherds and near complete vessels - collected between 2019 and 2023.

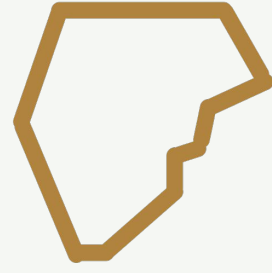
### WHY WAS IT CREATED?

To develop an extensive dataset of images of ceramic remains to train and test an AI system for the automatic identification and recording of pottery [1]

## THE DEPOSIT AT A GLANCE

28,160

SHERDS



384

NEAR-COMPLETE VESSELS



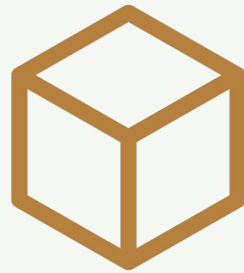
208,709

IMAGES



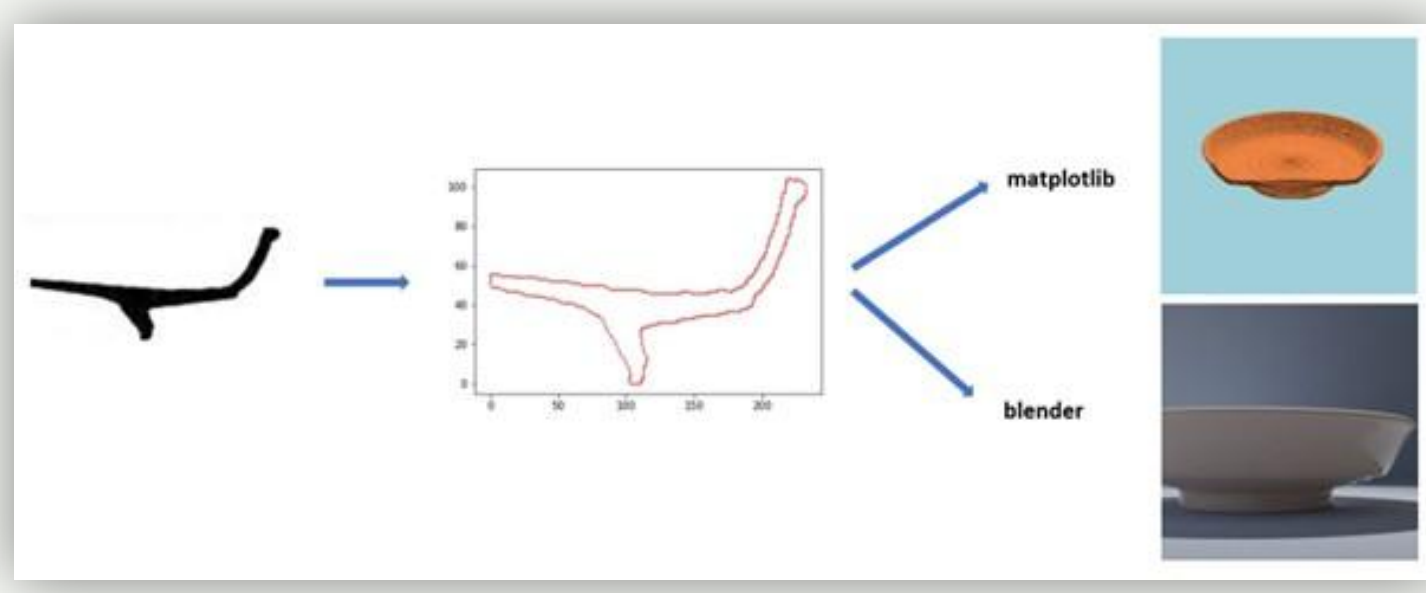
64

3D MODELS



698.78 GB

TOTAL DEPOSIT



## ARCHIVING A LARGE DATASET

The archive follows the Archaeology Data Service workflow, based on the Open Archival Information System (OAIS) reference model. The unusually large volume of data required several adaptations to the standard ingest process.

### 1 STAGED IMAGE MIGRATION

JPG>TIF

- Preservation copies migrated from .JPG to uncompressed .TIF
- Original .JPG files retained for dissemination
- Batch migration divided into stages and coordinated between two archivists and carried out with XnView MP

### 2 3D MODEL PRESERVATION

FBX>OBJ

- Deposited as proprietary .FBX
- Agreed to migrate them to object-specification .OBJ
- Blender used for conversion of untextured, low-poly models

### 3 METADATA & PROVENANCE

'is documented in'

- Images and 3D models associated with multiple metadata records
- Depositor documentation mapped to ADS metadata spreadsheet templates
- Preserved and created relationships between files and their descriptive metadata



SIP (Submission Information Package)



Processing (Migration + Metadata + Validation)



AIP (Archival Information Package)



DIP (Dissemination Information Package)

## CUSTOM QUERY INTERFACE

Our standard webpage template would have placed all 200,000+ images into a single tab with limited keyword filtering.

### Arch-I-Scan Search

Search this collection by choosing from the filters below or by looking for specific terms in the Title of each object in this collection. For example, you can use terms like *Samian* or *Decorated* to narrow your search.

**Data Type ?** ☐ 3D Models ☐ Images

**Item Type ?** ☒ Whole Vessels ☐ Shards

**Whole Vessels Form**

**Subject Keywords**

- ☐ beaker
- ☐ bowl
- ☐ carinated beaker
- ☐ carinated bowl
- ☐ cup
- ☐ dish
- ☐ flagon
- ☐ flanged bowl
- ☐ indented beaker
- ☐ inkwell
- ☐ jar
- ☐ miniature cup
- ☐ miniature jar
- ☐ miniature poppyhead beaker
- ☐ narrow necked jar
- ☐ ovoid beaker
- ☐ pedestal base
- ☐ plate
- ☐ poppyhead beaker
- ☐ poppyhead jar
- ☐ round bodied jar
- ☐ vase

**Title Search ?**

Records per page: ☒ 25 ☐ 50 ☐ 100

To address this, a bespoke search portal was created to provide advanced search parameters. [2]

The interface we created for Arch-I-Scan was adapted from a modular ADS template framework in place for special collections.

### Query Results

Your search:

- item type = whole\_vessels
- Subject keyword = cup

Has produced **3079** results.

Pages: 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | > >>

| Image | Description                                                                                                                               | More info                        |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|       | Standard photo image number 1 of L267+5, Highgate Wood 'C' ware miniature cup with grooves on exterior of vessel. Miniature of form file. | <a href="#">View full record</a> |
|       | Standard photo image number 2 of L267+5, Highgate Wood 'C' ware miniature cup with grooves on exterior of vessel. Miniature of form file. | <a href="#">View full record</a> |
|       | Standard photo image number 3 of L267+5, Highgate Wood 'C' ware miniature cup with grooves on exterior of vessel. Miniature of form file. | <a href="#">View full record</a> |
|       | Standard photo image number 4 of L267+5, Highgate Wood 'C' ware miniature cup with grooves on exterior of vessel. Miniature of form file. | <a href="#">View full record</a> |

### Query Results

Your search:

- Datatype = 3D Model

Has produced **64** results.

Pages: 1 | 2 | 3 | >

| Image | Description                             | More info                     |
|-------|-----------------------------------------|-------------------------------|
|       | A 3D model of terra sigillata form Cull | <a href="#">View 3D Model</a> |
|       | A 3D model of terra sigillata form Cull | <a href="#">View 3D Model</a> |
|       | A 3D model of terra sigillata form Cull | <a href="#">View 3D Model</a> |

### 3D Preview



1



### FILTER

Sherds / whole vessels / 3D models

Narrow results by data type, object type, ceramic form, and subject keywords

2



### SEARCH

Free-text title search

Find the desired result by using keywords, titles or IDs

3



### EXPLORE

Interactive 3D model viewer

Rotate, zoom, measure, adjust lighting and view 3D models in browser with 3DHOP



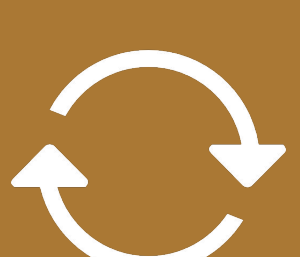
## FROM 698 GB OF DATA TO A REUSABLE ARCHIVAL WORKFLOW

Arch-I-Scan demonstrated that large scale datasets can be processed with our established digital preservation workflow - provided that ingest, metadata and discovery infrastructure are adapted to their scale



### STAGED PROCESSING

Large volume migration can be broken into manageable processing stages



### SUSTAINABLE DISCOVERY

Existing ADS special collection template can be adapted to support large-scale datasets without creating a bespoke system from scratch



### FUTURE-PROOFING

Lessons learnt from this experience help us inform on how to tackle similarly large deposits in the future

## References

- [1] D. van Helden, E. Mirkes, I. Tyukin, and P. Allison, "The Arch-I-Scan project: Artificial intelligence and 3D simulation for developing new approaches to Roman foodways," J. Comput. Appl. Archaeol., vol. 5, no. 1, pp. 78–95, 2022. DOI: 10.5334/jcaa.92
- [2] P. Allison and D. van Helden, "The Arch-I-Scan Project, 2019-2023," Archaeology Data Service [distributor], 2026. [Online]. DOI: 10.5284/1138130

## Acknowledgements

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